

Four-State Rydberg Dynamics

From Rabi Motion to Blockade Physics

Anchor: Two neutral atoms with basis $|gg\rangle, |gr\rangle, |rg\rangle, |rr\rangle$.

Purpose: Visualize how spacing, drive, detuning, and interaction suppress double excitation while allowing useful single excitation.

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Abstract. This classroom brief accompanies `sim_dynamics_Rydberg.ipynb`, a compact Jupyter notebook for visualizing two-atom Rydberg dynamics. The simulator evolves the four-state basis $|gg\rangle, |gr\rangle, |rg\rangle, |rr\rangle$ under a driven Hamiltonian with a spacing-dependent Rydberg interaction $V(R) = C_6/R^6$. Students use sliders to vary atom spacing, Rabi drive, detuning, total time, and time resolution. The goal is to see how population moves into the useful single-excitation manifold while double excitation is suppressed by the Rydberg blockade.

Introduction

The two-state Rabi simulator shows how a transverse drive moves population between two basis states. The Rydberg simulator adds the next ingredient needed for neutral-atom quantum technology: two driven atoms interact when both are excited to Rydberg states.

The four basis states are

$$|gg\rangle, \quad |gr\rangle, \quad |rg\rangle, \quad |rr\rangle.$$

The initial state is always

$$|\psi(0)\rangle = |gg\rangle.$$

The useful manifold is the single-excitation sector,

$$\{|gr\rangle, |rg\rangle\},$$

while the unwanted leakage channel is the double-excited state $|rr\rangle$.

Main observation.

$$P_{\text{single}} = P_{gr} + P_{rg} \text{ large} \quad \text{and} \quad P_{rr} \text{ small}$$

indicates useful excitation with suppressed double-excitation leakage.

Hamiltonian Used in the Simulator

In the ordered basis

$$|gg\rangle, |gr\rangle, |rg\rangle, |rr\rangle,$$

the simulator uses

$$H = \begin{pmatrix} 0 & \Omega/2 & \Omega/2 & 0 \\ \Omega/2 & -\Delta & 0 & \Omega/2 \\ \Omega/2 & 0 & -\Delta & \Omega/2 \\ 0 & \Omega/2 & \Omega/2 & -2\Delta + V(R) \end{pmatrix}.$$

The spacing-dependent interaction is

$$V(R) = \frac{C_6}{R^6}.$$

The notebook uses the rubidium-scale value

$$\frac{C_6}{2\pi} = 862690 \text{ MHz } \mu\text{m}^6.$$

The sliders use

$$R \text{ in } \mu\text{m}, \quad \Omega/2\pi, \Delta/2\pi, V/2\pi \text{ in MHz}, \quad t \text{ in } \mu\text{s}.$$

The state evolves according to

$$|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle.$$

The simulator then computes

$$P_{gg}(t), \quad P_{gr}(t), \quad P_{rg}(t), \quad P_{rr}(t),$$

and checks

$$P_{gg} + P_{gr} + P_{rg} + P_{rr} = 1.$$

Guided Student Exercises

Exercise 1: Check probability conservation.

Run the simulator using the default settings.

Prompts.

1. Does the total probability remain equal to one?
2. What is the maximum normalization error?
3. Why is this a useful first check before interpreting the physics?

Exercise 2: Identify the useful and unwanted populations.

Inspect the four-state probability plot and the single-excitation/leakage plot.

Prompts.

1. Which states form the useful single-excitation manifold?
2. Which state corresponds to double-excitation leakage?
3. Why is P_{gg} not the main performance metric?

Exercise 3: Strong blockade at short spacing.

Set

$$R = 5.0 \mu\text{m}, \quad \Omega/2\pi = 0.318 \text{ MHz}, \quad \Delta/2\pi = 0.$$

Prompts.

1. Is P_{rr} strongly suppressed?
2. Does population enter the single-excitation manifold?
3. What does the large value of $V(R)/2\pi$ tell you?

Exercise 4: Weaken the blockade by increasing spacing.

Increase R from $5 \mu\text{m}$ to $7, 8, 9$, and $10 \mu\text{m}$, leaving Ω , Δ , and total time fixed.

Prompts.

1. How does $V(R)/2\pi$ change as R increases?
2. When does P_{rr} become visibly nonzero?
3. What does this show about the R^{-6} interaction?

Exercise 5: Explore pulse timing.

Choose a fixed spacing, for example $R = 7 \mu\text{m}$, and vary the total time.

Prompts.

1. Does the final distribution depend on when the system is measured?
2. Can P_{gg} become large again after population has entered the single-excitation manifold?
3. Why is the simulator showing dynamics rather than a static blockade rule?

Exercise 6: Optional detuning exploration.

Keep R and Ω fixed and vary Δ .

Prompts.

1. Does detuning change how easily population enters the single-excitation manifold?
2. Does detuning change the apparent leakage into $|rr\rangle$?
3. Why is $\Delta = 0$ the cleanest starting point for the basic blockade exercise?

Takeaway message. The Rydberg blockade is not a classical rule that forbids $|rr\rangle$. It is a quantum suppression mechanism: the interaction shift $V(R)$ moves the double-excited state away from resonance. Shorter spacing gives stronger interaction, stronger blockade, and lower double-excitation leakage. The useful operating behavior is large single-excitation probability together with small P_{rr} .

Sample Answers to Prompts

Exercise 1. The total probability should remain equal to one within numerical precision. A small normalization error, typically near floating-point precision, indicates that the matrix-exponential time evolution is behaving as a unitary evolution.

Exercise 2. The useful single-excitation states are $|gr\rangle$ and $|rg\rangle$. The leakage state is $|rr\rangle$. The state $|gg\rangle$ is the known starting point and can also reappear due to coherent oscillation, but the main performance quantities are P_{single} and P_{rr} .

Exercise 3. At short spacing, $V(R)$ is large, so $|rr\rangle$ is shifted away from resonance. The simulator should show suppressed P_{rr} and visible population in the single-excitation manifold.

Exercise 4. As R increases, $V(R)$ decreases rapidly because $V(R) \propto R^{-6}$. Once the interaction shift becomes less dominant compared with the drive, P_{rr} becomes more visible. This illustrates that blockade is a crossover, not a hard threshold.

Exercise 5. The final distribution depends strongly on the measurement time. Population can move from $|gg\rangle$ into the single-excitation manifold and later return toward $|gg\rangle$. This is coherent quantum dynamics, not a static probability assignment.

Exercise 6. Detuning changes which states are energetically accessible to the drive. For the basic blockade exercise, $\Delta = 0$ is cleanest because the suppression of $|rr\rangle$ is then mainly due to the interaction shift $V(R)$.

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